Automotive Black Box (EDR)

with Wireless Vehicle Control System

A Mini Project report submitted of the requirements for the degree of

BACHELOR OF TECHNOLOGY

IN

ELECTRONICS AND COMMUNICATION ENGINEERING

by

A. LAKSHMI SAI GANESH	S210732
M. SUJEEV KUMAR	S210799
V. PAPA	S210592
K. VASANTHA	S210120
B. MADHURIMA	S210802

Under the esteemed Guidance of

Dr H SRINIVASA VARAPRASAD, M.tech, Ph.D

Assistant professor

Dept of ECE

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

RAJIV GANDHI UNIVERSITY OF KNOWLEDGE TECHNOLOGIES

SRIKAKULAM(DIST-532410)

APRIL-2026

CERTIFICATE

This is to certify that the work entitled "**Automotive Black Box (EDR) with Wireless Vehicle Control System**" is a Bonafide record of authentic work carried out by A. Lakshmi Sai Ganesh-S210732, M. Sujeev Kumar-S210799, V.Papa-S210592 ,K.Vasantha-S210120, B.Madhurima-S210802 under my supervision and guidance for the fulfilment of the requirement of the award of the degree of Bachelor of Technology in the department of Electronics and Communication on Engineering at RGUKT-SRIKAKULAM.

The results embodied in this work have not been submitted to any other university or institute for the award of any degree or diploma.

Project Supervisor
Dr H Srinivasa Varaprasad , M.Tech, Ph.D
Assistant Professor
ECE Department,
RGUKT-Srikakulam

Head of the department
T S GAGANDEEP
Head of the Department ECE,
ECE Department,
RGUKT-Srikakulam

DECLARATION

We hereby declare that the dissertation entitled **Automotive Black Box (EDR) with Wireless Vehicle Control System** submitted for the B.Tech Degree is our original work and the dissertation has not formed the basis for the award of any degree, associateship, fellowship or any other similar titles.

PLACE: Srikakulam
DATE: 2026-04-23

A. Lakshmi Sai Ganesh (S210732)

M. Sujeev Kumar (S210799)

V. Papa (S210592)

K. Vasantha (S210120)

B. Madhurima (S210802)

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to the following individuals and organizations for their invaluable support and contributions to this project **“Automotive Black Box (EDR) with Wireless Vehicle Control System”**.

Guides and Mentors:

- Dr.H.Srinivasa Varaprasad for their guidance and valuable insights.

ELECTRONICS AND COMMUNICATION ENGINEERING SRIKAKULAM

DATE: 2026-04-23

PLACE: SRIKAKULAM

ABSTRACT

Event Data Recorders (EDRs), commonly known as automotive black boxes, are critical systems used for accident analysis, safety monitoring, and vehicle diagnostics. However, commercial EDR systems are expensive, proprietary, and integrated into high-end vehicle control units, making them unsuitable for academic and low-cost implementations.

This project presents a **low-cost embedded automotive black box prototype** using an ESP32 microcontroller and multiple sensors to monitor temperature, vibration, and acceleration. The system implements a **ring buffer mechanism** to store approximately 3 minutes of pre-crash data, which is automatically written to an SD card upon crash detection.

Crash detection is achieved using a **dual-condition logic** based on vibration and g-force thresholds, improving reliability over single-sensor systems. The system also provides **real-time monitoring through ThingsBoard cloud (via MQTT)** and a **local web dashboard hosted on ESP32**, enabling live visualization of sensor data.

Additionally, the prototype supports **wireless vehicle control**, demonstrating integration between safety monitoring and control systems. The proposed design offers a **cost-effective, scalable, and educational platform** aligned with core EDR principles.

TABLE OF CONTENTS

1.INTRODUCTION

1.1 Background

In modern automotive systems, safety and data analysis have become increasingly important due to the rising number of road accidents and the need for accurate fault analysis. Event Data Recorders (EDRs), often called "automotive black boxes," are designed to record vehicle parameters before, during, and after a crash event. These systems provide valuable data such as acceleration, braking behavior, and system status, which can be used for accident reconstruction and safety improvements.

Traditional EDR systems are typically embedded within vehicle control units such as Airbag Control Modules (ACMs) and are governed by industry standards like IEEE 1623. While these systems are highly reliable, they are expensive and not accessible for educational or experimental purposes.

With advancements in embedded systems, IoT, and microcontroller technologies, it has become feasible to design **low-cost EDR prototypes** using platforms like ESP32. These systems allow students and researchers to understand the working principles of black box systems without requiring expensive hardware.

This project aims to replicate the fundamental concepts of an automotive black box using affordable components while maintaining essential features such as **crash detection, data logging, and real-time monitoring**.

1.2 Problem Statement

Despite the importance of EDR systems, there are several limitations in existing solutions:

- Commercial EDR systems are **costly and proprietary**
- Low-cost prototypes often lack:
  - Pre-crash data storage
  - Multi-sensor integration
  - Real-time monitoring capabilities
- Many systems rely on **single-sensor detection**, leading to false alarms
- Lack of integration between **monitoring and control systems**

These limitations create a gap between theoretical understanding and practical implementation of automotive black box systems.

Therefore, the problem is to design a system that:

- Is **low-cost and accessible**
- Supports **multi-sensor data acquisition**
- Implements **reliable crash detection**
- Provides **pre-crash data logging**
- Enables **real-time monitoring and control**

1.3 Objectives

The primary objectives of this project are:

- To design a low-cost embedded automotive black box using ESP32
- To acquire real-time data from multiple sensors (temperature, vibration, acceleration)
- To implement a **dual-condition crash detection mechanism**
- To develop a **ring buffer system** for storing pre-crash data
- To log system data into an SD card in CSV format
- To enable **real-time cloud monitoring using ThingsBoard**
- To create a **local web dashboard for live visualization and control**
- To integrate **wireless vehicle control features**.

1.4 Scope of the Project

This project focuses on developing a prototype-level EDR system suitable for academic and demonstration purposes. The scope includes:

- Real-time sensor data acquisition
- Crash detection using threshold-based logic
- Pre-crash and post-crash data logging
- IoT-based monitoring
- Local web-based interface

The system does not include:

- Industrial-grade automotive sensors
- Full vehicle integration

- Advanced security mechanisms

However, it provides a strong foundation for further development.

1.5 Organization of the Report

- **Chapter 1: Introduction**
- **Chapter 2: Literature Survey**
- **Chapter 3: System Architecture & Methodology**
- **Chapter 4: System Design & Implementation**
- **Chapter 5: Results & Analysis**
- **Conclusion & Future Scope**

2.LITERATURE SURVEY

Event Data Recorders have evolved significantly over the years, from simple data logging devices to advanced systems integrated into modern vehicles.

Early EDR systems focused on recording basic parameters such as speed and acceleration. However, with the advancement of embedded systems and IoT, modern systems now include multiple sensors, cloud connectivity, and intelligent event detection mechanisms.

Several research works highlight different approaches:

- **Low-cost EDR systems** use microcontrollers and accelerometers but lack advanced features
- **Smart Black Box systems (IEEE)** use intelligent buffering but require high computational power
- **IoT-based systems** enable remote monitoring but depend on network availability
- **Blockchain-based systems** improve security but are complex and resource-intensive

Limitations of Existing Systems

- High cost and complexity
- Lack of pre-crash data logging
- Limited real-time monitoring
- Dependence on single-sensor detection
- Poor scalability for low-cost applications

Research Gap

There is a need for a system that:

- Combines low cost + multi-sensor logging + real-time monitoring
- Implements pre-crash data storage
- Provides both local and cloud-based visualization

Event Data Recorder / EDR / Automotive Black Box:

S. No .	Year	Title / Authors	Methodology	Work Done	Limitations
1	2018	"The Smart Black Box: A Value-Driven Automotive Event Data Recorder" – Yu Yao, Ella Atkins (IEEE-ITSC)	Uses a value-driven Mealy-machine-based buffer to selectively store high-bandwidth sensor data (video, object-detection outputs) only during anomalies or high-risk events; combines with low-bandwidth CAN-like logs.	Designs a Smart Black Box (SBB) that augments conventional low-bandwidth EDRs with event-triggered, compressed high-bandwidth data (e.g., camera-like streams) for autonomous-vehicle validation and accident analysis.	Focused on autonomous/high-end vehicles; computationally heavy; not optimized for low-cost microcontroller-based academic prototypes.
2	2013	"Low Cost Black Box for Cars" – Sumam David et al.	Implements a low-cost embedded system using microcontroller, accelerometer, and display units to record speed and acceleration directly on the device.	Builds a domestic-vehicle black-box that records real-time speed and acceleration and reduces cost by reusing the same hardware for multiple parameters.	Limited to basic CAN-style data (speed, acceleration); no wireless or cloud-based transmission; not suitable for advanced IoT/remote-inspection features.
3	2017	"Event Data Recorder for Investigation, Legal Claim and Fault Analysis in Vehicles" – Dhanam, Marichamy, Mohan et al.	Uses microcontroller + accelerometer + GPS to record vehicle dynamics and location; triggers logging and alerting using threshold-based abnormal-event detection logic.	Develops a low-budget EDR for accident investigation, legal claims, and fault diagnosis, storing pre- and post-crash data and supporting basic forensic analysis.	Lacks advanced security/privacy mechanisms; logging resolution and crash reliability unclear for high-speed scenarios.
4	2021	"Affordable Black Box: A Smart Accident Detection System for Cars" – IEEE	Employs embedded sensors (accelerometer, GPS) and a threshold-based accident-detection algorithm; logs data locally and triggers	Presents a cost-effective accident-detection and black-box system targeting low-budget and educational/prototype applications.	Mainly focuses on basic crash detection; lacks detailed safety monitoring (temperature, vibration, driver-state) and standardized EDR

			alerts/transmission.		parameter definitions.
5	2021	"Secure, Privacy-Preserving, Distributed Motor Vehicle Event Data" – IEEE	Uses blockchain-based distributed storage and encryption to record vehicle-incident data in a tamper-resistant, privacy-preserving manner.	Designs a secure, decentralized EDR framework for smart-city vehicles where incident data are stored immutably and accessed only with proper authorization.	Requires complex infrastructure (blockchain nodes, cloud); high overhead and latency; not suitable for small-scale low-cost embedded prototypes.
6	2005/2008	"IEEE Standard for Motor Vehicle Event Data Recorder (MVEDR)" – IEEE Std 1623	Defines standardized data elements, export protocols, and functional requirements for MVEDRs including connector-lockout and data-compatibility rules.	Provides a formal standard for what data should be collected, how it should be stored, and how it can be exported, forming the baseline for commercial EDRs.	High-level regulatory standard; not an implementable hardware/software prototype; hardware-agnostic and not directly usable in embedded lab designs.
7	2014	"THE USE OF EVENT DATA RECORDER (EDR) – BLACK BOX" – ASTJ	Survey-style analysis discussing EDR applications in courts, insurance, and regulation based on existing standards and case-law.	Explains how EDRs act as a neutral witness in accident investigations, legal claims, and insurance-fraud analysis.	Not a technical implementation; no hardware/software design details; mainly legal/policy-oriented.
8	2025	"Vehicle Black Box System" – IJSRT	Uses IoT-enabled microcontroller (e.g., ESP8266) with accelerometer, gyroscope, GPS, and Wi-Fi to monitor vehicle dynamics and trigger emergency alerts on crash detection.	Implements a low-cost IoT-based black-box that records motion, tilt, and location, sends crash alerts with GPS coordinates, and stores data locally and in the cloud.	Relies on cloud/Wi-Fi connectivity; may be unreliable in low-network areas; security and privacy mechanisms are basic.

Collision / Accident Detection and Safety-Monitoring works:

S. No.	Year	Title / Authors	Methodology	Work Done	Limitations
1	2019	“Wireless Bluetooth Car Collision Detection System” – Yuying Wang, Kaixuan Hu, Ya Zhou (IEEE)	Uses Bluetooth-enabled wireless communication between vehicles combined with onboard sensors to detect proximity and broadcast collision-risk warnings. Relies on peer-to-peer short-range Bluetooth communication between cars.	Proposes a wireless collision-detection system where vehicles detect nearby cars wirelessly via Bluetooth and generate early-warning alerts to avoid collisions. Suitable as a reference for Bluetooth-controlled safety-aware vehicles.	Requires multiple vehicles equipped with compatible Bluetooth modules; suitable mainly for dedicated short-range scenarios; not optimized for single robotic-car or low-cost academic setups.
2	2023	“Ultrasonic Sensor-Based Embedded System for Vehicular Collision Detection and Alert” – SCIRP Journal	Uses multiple ultrasonic sensors with an Arduino-based microcontroller to detect front and rear obstacles; calculates distance from echo-time and triggers buzzer/LCD alerts when thresholds are crossed.	Implements a low-cost embedded collision-detection and alert system providing audible buzzer warnings and LCD distance display; supports both forward and backward obstacle detection.	Does not integrate automatic braking or speed control; only provides warnings. Ultrasonic sensors are sensitive to angle, noise, and environmental conditions.
3	2017	“Embedded Based Vehicle Speed Moderation and Collisionless System” – IJRTI (ECE-Udupi, India)	Uses Arduino-Uno, ultrasonic sensors, and Bluetooth module to measure obstacle distance and automatically reduce speed	Develops an Arduino-based collision-avoidance and speed-moderation system where the vehicle slows down or stops before	Lacks formal data logging or EDR-style recording; focuses only on real-time control and feedback; not suitable as standalone

			via DC motor control; also monitors temperature and fuel level.	collision; supports smartphone/PC remote control via Bluetooth.	black-box for accident analysis.
4	2019	“IoT-Based Non-Intrusive Automated Driver Drowsiness Monitoring Framework” – M. A. Khan, M. Nawaz et al.	Uses IoT-based sensing including camera and OBD-II vehicle data to monitor eye-closure, head-pose, and vehicle dynamics; triggers alerts when drowsiness is detected.	Proposes a drowsiness-detection framework for logistics and public-transport vehicles integrating camera and OBD data in an IoT platform to improve road safety.	Focused on driver-state monitoring rather than collision detection or event-data recording; requires camera and backend cloud processing; heavy for low-cost microcontroller prototypes.
5	2019	“Voice and Bluetooth Controlled Robot Car” – IRJET	Uses Android voice-control app to send Bluetooth commands to an Arduino-controlled robot car with DC motors and basic obstacle-avoidance logic.	Builds a robotic vehicle controllable via voice or mobile-app buttons using Bluetooth communication; supports obstacle avoidance and remote navigation.	Focuses on control and navigation; minimal safety-monitoring or event-data recording capability; limited sensor integration.

IoT / Bluetooth-controlled robotic-vehicle works:

S. No.	Year	Title / Authors	Methodology	Work Done	Limitations
1	2023	“Design and Implementation of an IoT-Enabled Bluetooth Robot Car with Obstacle Avoidance” – B. Swathi, S. Kamsali, K. Gayathri, M. D. Lakshmi (IJISAE)	Uses an IoT-enabled Bluetooth robot car with two-way Bluetooth communication between a mobile-based controller and the robot; implements a PD-based	Builds a Bluetooth-controlled robot car capable of autonomous obstacle avoidance within a defined area; optimizes sensor-reading frequency and	Focused on navigation and obstacle avoidance; does not implement systematic event-data recording or black-box-style logging; lacks crash-data storage

			obstacle-avoidance algorithm with continuous sensor-data transmission.	communication data-rate to balance real-time response and reliability.	for post-incident analysis.
2	2017	"Embedded Based Vehicle Speed Moderation and Collisionless System" – ECE-Udupi, IJRTI	Uses Arduino-Uno, ultrasonic sensors, and Bluetooth module to measure obstacle distance and automatically reduce vehicle speed; monitors temperature and fuel level; supports smartphone/PC control via Bluetooth.	Implements a collision-avoidance and speed-moderation system where the vehicle slows down or stops before collision; enables Bluetooth-based remote control and real-time monitoring.	Does not support event-data logging or EDR-style black-box functionality; provides only real-time control and feedback; unsuitable for accident analysis or forensic investigation.
3	2018	"Black Box with Intelligent Vehicle System" – Prafulla U. P., Yuvashree G. S., Ramya Gowda (IJERT)	Uses PIC16F877A microcontroller with EEPROM to store 21 seconds of pre-accident and 10 seconds of post-accident data (speed, brake status, lights, etc.); data accessed via serial PC interface using VB.NET software.	Implements a vehicle black-box system that continuously records vehicle parameters and securely stores crash-related data retrievable only after an accident.	Requires PC-based VB.NET interface for data retrieval; lacks wireless/IoT connectivity; less suitable for fully embedded or cloud-integrated academic prototypes.
4	2019	"Voice and Bluetooth Controlled Robot Car" – IRJET	Uses an Android voice-control application to send Bluetooth commands to an Arduino-controlled robot car with four DC	Develops a Bluetooth-based voice-controlled robot car supporting user-friendly navigation and basic obstacle avoidance	Primarily focused on control and usability; does not include safety monitoring, event-data recording, or EDR-style

			motors; voice/button inputs converted into digital control strings.	through RF communication.	logging; limited sensor integration beyond motion control.

Heavier-Duty / Automotive EDR Standards and Surveys:

S. No.	Year	Title / Authors	Methodology	Work Done	Limitations
1	2024	“Event Data Recorder: The Black Box for Vehicles – DEKRA” – Michael Vogel (DEKRA Technical Overview)	Survey-style technical description of modern automotive EDR systems, explaining recorded parameters, triggering mechanisms, and physical integration (typically within the airbag control unit).	Describes how EDRs continuously monitor longitudinal/lateral speed, accelerator and brake position, engine speed, steering angle, ABS/ESC status, seat-belt usage, and airbag-related events, storing data only when crash-like acceleration thresholds are detected.	Not a formal academic paper or technical standard; descriptive and policy-oriented; lacks hardware/software implementation details for direct circuit design use.
2	2007	“Use of Event Data Recorder (EDR) Technology for Highway Safety Research / NCHRP Report W75” – TRB (Transportation Research Board)	Survey-based evaluation of EDR usage in highway safety research and legal contexts; discusses recommended data elements, extraction procedures, and privacy considerations.	Explains how EDRs provide objective crash-related data (speed, acceleration-time history, braking, seat-belt status, airbag deployment) for accident reconstruction, safety research, and litigation support.	High-level and policy-focused; no implementation-level technical design; oriented toward regulatory and large-scale safety research rather than embedded prototypes.
3	2018	“The Smart Black Box: A Value-Driven Automotive Event Data Recorder” – Yu Yao, Ella Atkins (IEEE ITSC / IEEE T-ITS)	Uses a value-driven Mealy-machine-based buffer architecture to selectively store high-bandwidth data (camera-	Introduces the Smart Black Box (SBB) extending conventional EDRs with intelligent high-bandwidth event capture for autonomous-vehicle validation	Designed for advanced autonomous vehicles with significant computational resources; too complex and resource-intensive

			like streams, object-detection outputs) during risk-relevant events, alongside low-bandwidth CAN-style logs.	and detailed accident reconstruction.	for low-cost Arduino or small MCU-based academic prototypes.
4	2021	"Towards Secure, Decentralised, and Privacy-Friendly EDR-like Framework" – IEEE (Blockchain-Based Architecture)	Employs blockchain-based decentralized storage where vehicular incident data are encrypted, distributed across nodes, and accessed only through authorized mechanisms.	Provides a secure and tamper-resistant architecture for recording vehicular incident data, enabling privacy preservation and reliable liability attribution.	Requires high-end connectivity and blockchain infrastructure; introduces latency and system complexity; unsuitable for low-cost embedded or academic-level implementations.

Sensor-Based Monitoring and Embedded-System Works:

S. No.	Year	Title / Authors	Methodology	Work Done	Limitations
1	2023	"Ultrasonic Sensor-Based Embedded System for Vehicular Collision Detection and Alert" – SCIRP Journal	Uses multiple ultrasonic sensors with an Arduino-based microcontroller to measure front and rear obstacle distance; applies a threshold-based proximity algorithm to trigger a buzzer and LCD alert simulating collision detection.	Implements a compact, low-cost embedded collision-detection system that provides audible warnings and visual distance display; uses a distance-based decision algorithm to activate alerts.	Provides only driver-assist alerts without automatic braking or speed control; does not store structured event data for post-crash analysis; not suitable as an EDR-style black-box.
2	2023	"IoT-Based Non-Intrusive Automated	Uses camera-based visual features (eye-	Proposes an end-to-end IoT-based	Designed for high-end or commercial

		Driver Drowsiness Monitoring Framework" – M. A. Khan, M. Nawaz et al. (IEEE IoT Framework)	closure detection, head-pose estimation) combined with IoT-enabled transmission and OBD-II-like vehicle dynamics data to detect drowsiness and generate alerts.	driver-monitoring framework integrating sensors, cloud processing, and alert mechanisms to improve safety in logistics and public-transport vehicles.	vehicles requiring cameras and backend servers; computationally intensive; unsuitable for low-cost microcontroller-based academic prototypes.
3	2017	"Embedded Based Vehicle Speed Moderation and Collisionless System" – ECE-Udupi, IJRTI	Uses Arduino-Uno, ultrasonic sensor, Bluetooth module, DC motor, and temperature/fuel sensors to detect obstacles, reduce speed, and optionally stop before collision; supports smartphone/PC control via Bluetooth.	Develops an Arduino-based system that performs obstacle detection, automatic speed moderation, halt-before-collision control, and real-time monitoring through Bluetooth access.	Focuses on real-time control rather than systematic event logging; lacks EDR-style data recording and structured parameter storage for accident reconstruction.
4	2023	"Design and Implementation of an IoT-Enabled Bluetooth Robot Car with Obstacle Avoidance" – B. Swathi, S. Kamsali, K. Gayathri, M. D. Lakshmi (IJISAE)	Uses two-way Bluetooth communication and a PD-controller-based obstacle-avoidance algorithm; dynamically adjusts sensor-reading frequency and communication rate for stable control.	Implements a Bluetooth-controlled robot car capable of autonomous obstacle avoidance within a defined environment; demonstrates IoT integration with robotics for reliable sensor-control loops.	Emphasis is on navigation and communication reliability; does not include safety monitoring, EDR-style logging, or structured multi-parameter recording for later analysis.

3.SYSTEM ARCHITECTURE AND METHODOLOGY

3.1 System Architecture

The proposed system is designed as a **modular embedded architecture** centered around the ESP32 microcontroller, which acts as the brain of the entire system. It performs data acquisition, processing, decision-making, storage, communication, and control operations.

The architecture integrates multiple hardware and software components to achieve a reliable and real-time automotive black box system.

Core Components and Their Functions

- **ESP32 Microcontroller**
    - Acts as the central controller
    - Handles sensor data processing and decision-making
    - Manages communication with cloud and web dashboard
- **Temperature Sensor (Analog Input)**
    - Measures ambient/system temperature
    - Used for monitoring abnormal thermal conditions
- **Vibration Sensor (Digital Input)**
    - Detects physical shocks or impacts
    - Used as a primary indicator for collision detection
- **MPU6050 Accelerometer (I2C Interface)**
    - Provides 3-axis acceleration data
    - Used to calculate resultant g-force
    - Helps detect sudden motion changes
- **SD Card Module (SPI Interface)**
    - Stores continuous and crash-event data
    - Maintains CSV logs for analysis
- **Wi-Fi Module (ESP32 Built-in)**

- Enables communication with ThingsBoard cloud
- Sends telemetry using MQTT protocol

- **Local Web Server**
    - Hosted on ESP32
    - Displays real-time data and control interface
- **Motor Driver (L298N)**
    - Controls DC motors for vehicle simulation
    - Demonstrates safety response during crash
- **Alert LED / Alarm Pin**
    - Indicates crash detection visually

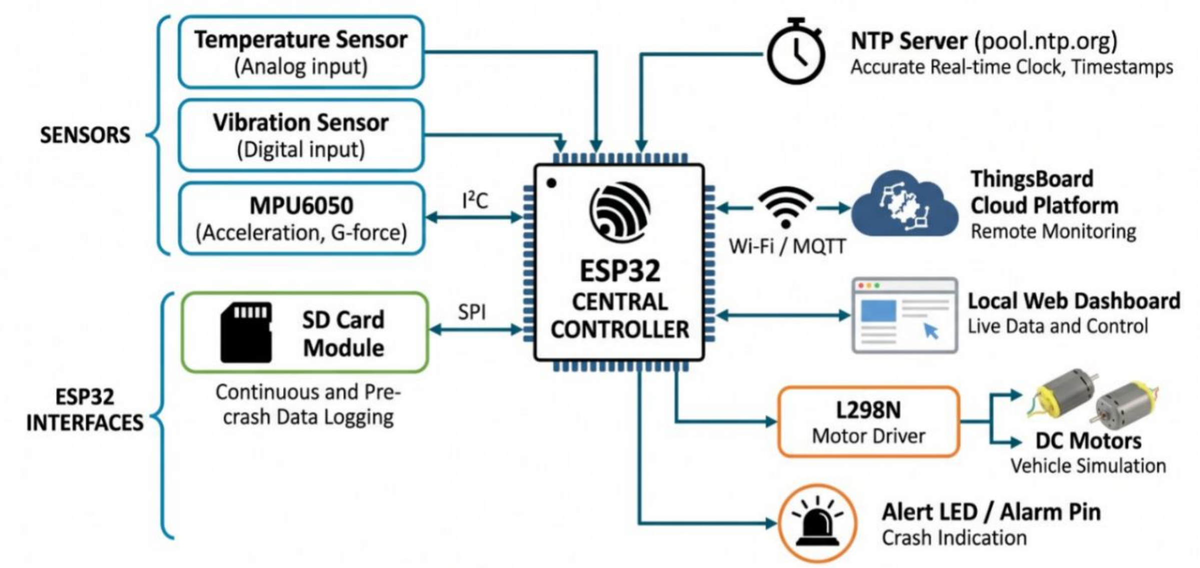

Fig 3.1: System Architecture of Mini Automotive Black Box

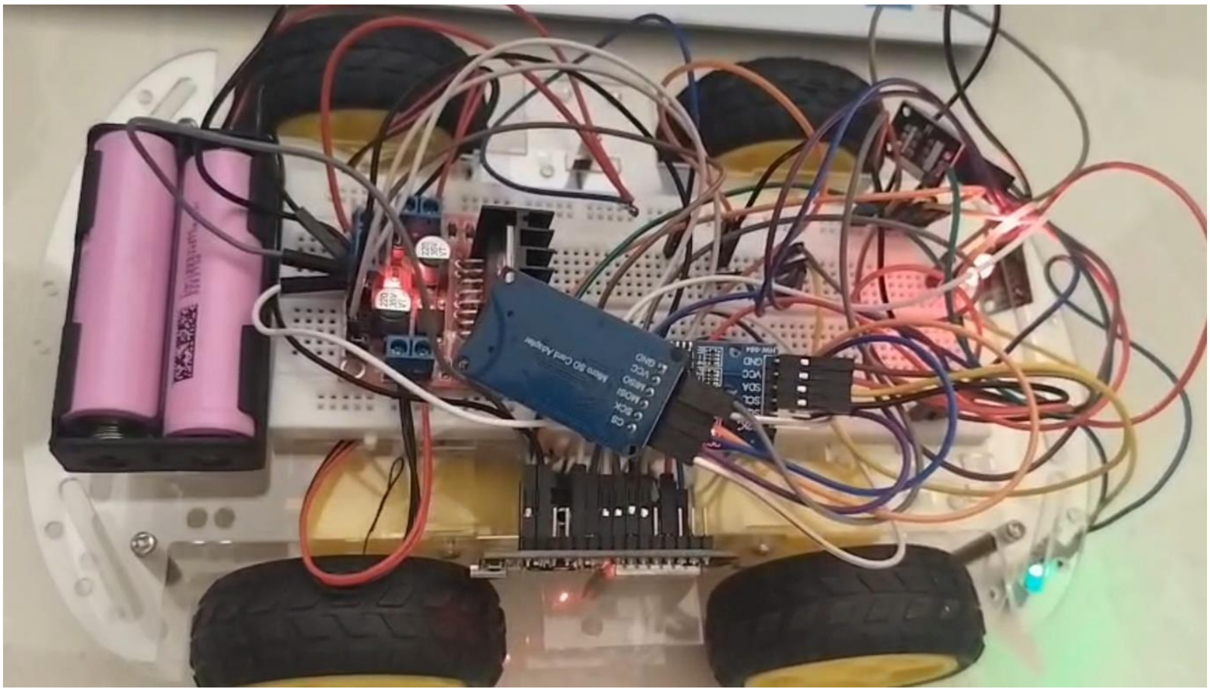

Fig 3.2: Hardware Implementation of ESP32-Based Automotive Black Box System

3.2 Methodology

The system operates through a structured sequence of steps to ensure continuous monitoring and reliable crash detection.

Step 1: Data Acquisition

The ESP32 continuously collects data from multiple sensors in real-time:

- Temperature values from analog sensor
- Vibration pulses from digital sensor
- Acceleration data from MPU6050

This multi-sensor approach ensures better accuracy compared to single-sensor systems.

Step 2: Ring Buffer Storage

A fixed-size circular buffer is implemented in ESP32 RAM to store recent data.

- Stores approximately 3 minutes of data
- Each entry includes:

    - Timestamp
    - Temperature
    - Vibration
    - g-force
    - System status
- When buffer is full:
    - Oldest data is overwritten automatically

This ensures that pre-crash data is always available.

Step 3: Crash Detection

The system uses a dual-condition detection logic:

- Crash is detected if:
    - Vibration ≥ threshold OR
    - g-force exceeds predefined limit

This method:

- Reduces false alarms
- Improves reliability

Step 4: Crash Response

Once a crash is detected:

- System status changes to "CRASH DETECTED"
- Alert LED turns ON
- Motors are immediately stopped
- Pre-crash buffer data is prepared for storage

Step 5: Data Logging

Data is continuously written to the SD card:

- Format: CSV
- Each record contains:
    - Time
    - Temperature
    - Vibration
    - g-force
    - Status

During crash:

- Special markers are added:
    - “PRE-CRASH LOG”
    - “CRASH EVENT”

Step 6: Real-Time Monitoring

The system supports dual monitoring:

1. Cloud Monitoring (ThingsBoard)
    - Data sent via MQTT
    - Enables remote visualization
2. Local Web Dashboard
    - ESP32 hosts a web page
    - Updates every second

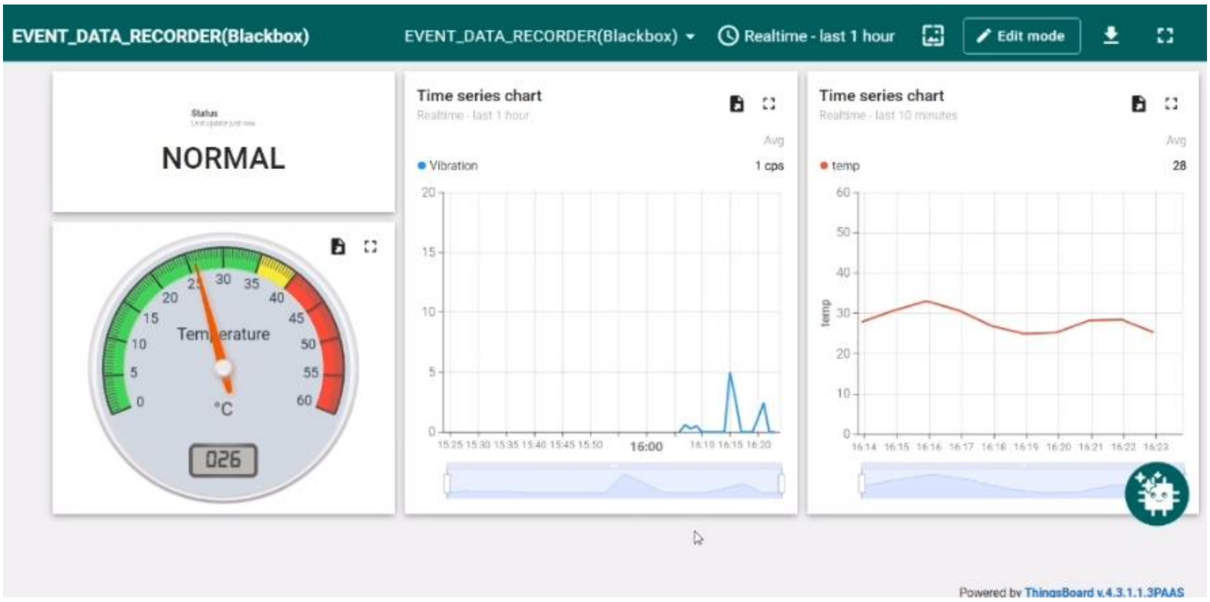

Figure 3.3: ThingsBoard Cloud Dashboard Showing Real-Time Data

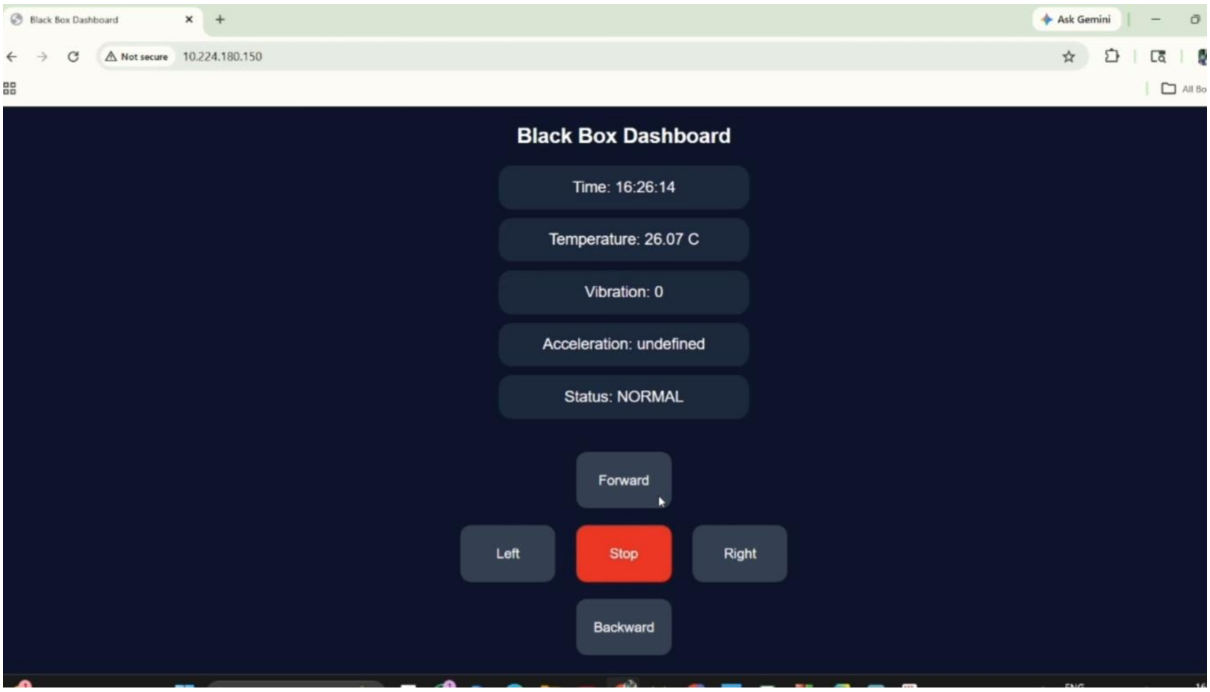

Figure 3.4: Local Web Dashboard Showing Normal System Status

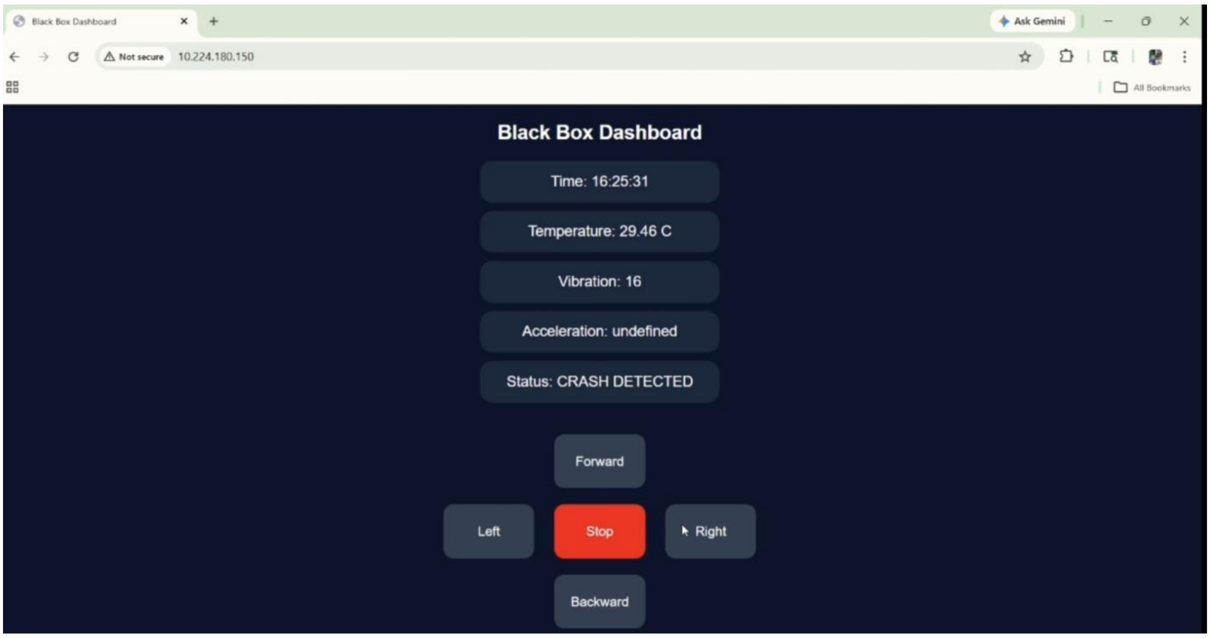

Figure 3.5: Local Web Dashboard Showing Crash Detection Status

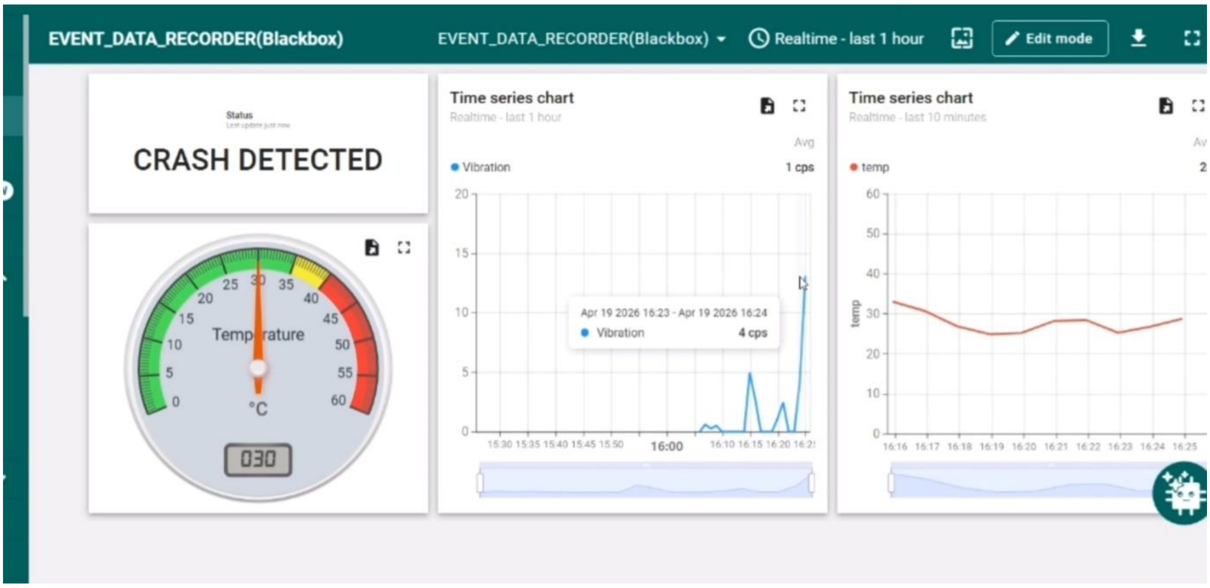

Figure 3.5: ThingsBoard Dashboard Showing Crash Event Detection

Step 7: Auto Restart Mechanism

- After crash detection:
  - System waits for predefined delay (~2 minutes)
  - Automatically restarts using ESP.restart()

This ensures:

- System recovery
- Continuous monitoring

3.3 Technologies Used

The implementation of the Automotive Black Box system involves a combination of hardware, software, and communication technologies as listed below:

Hardware Components

- ESP32 Microcontroller (central processing unit)
- MPU6050 (Accelerometer for g-force detection)
- Vibration Sensor (impact detection)
- Temperature Sensor (analog input)
- SD Card Module (data logging)
- L298N Motor Driver (vehicle simulation)
- DC Motors and Alert LED

Software Tools

- Arduino IDE for programming ESP32
- Embedded C/C++ for firmware development
- HTML, CSS, and JavaScript for web dashboard

Communication Technologies

- Wi-Fi for wireless connectivity
- MQTT protocol for real-time cloud communication
- HTTP protocol for local web server

Cloud Platform

- ThingsBoard IoT platform for telemetry visualization and monitoring

Time Synchronization

- NTP (Network Time Protocol) for accurate timestamping

4.SYSTEM DESIGN AND IMPLEMENTATION

4.1 System Design

The system is designed using a **modular approach** for better scalability and maintainability.

Modules:

1. **Sensor Module**
   - Collects environmental and motion data
2. **Processing Module**
   - Processes sensor inputs
   - Executes crash detection logic
3. **Storage Module**
   - Manages SD card logging
   - Handles ring buffer
4. **Communication Module**
   - Handles Wi-Fi and MQTT
   - Sends data to cloud
5. **Control Module**
   - Controls motors
   - Handles user commands

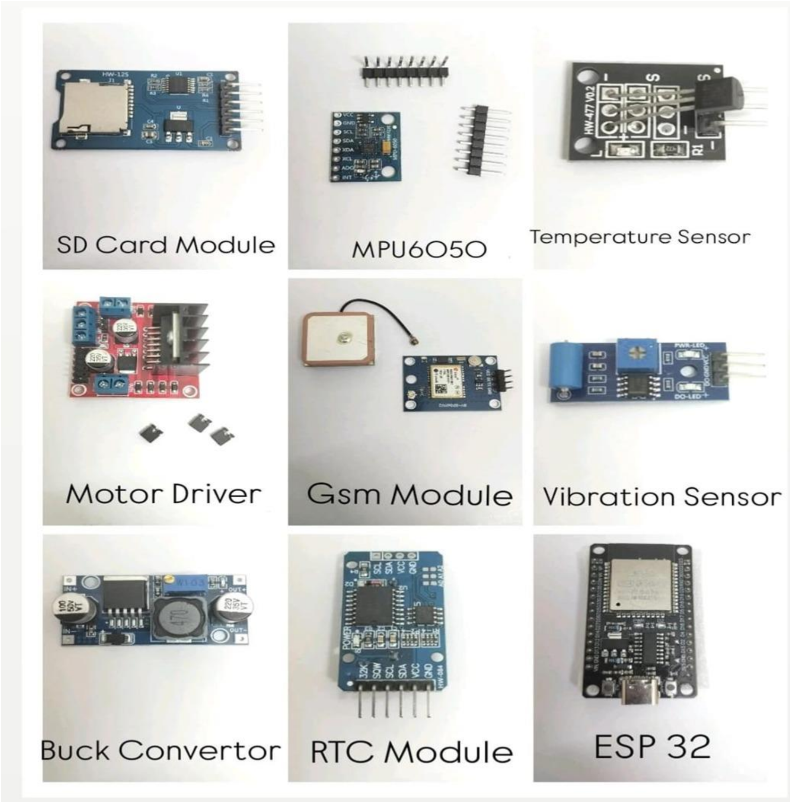

Figure 4.1: Hardware Components

4.2 Implementation Details

- Microcontroller: ESP32
- Programming: Embedded C (Arduino IDE)
- Protocol: MQTT
- Cloud Platform: ThingsBoard
- Storage: SD Card (CSV format)
- Time Sync: NTP Server

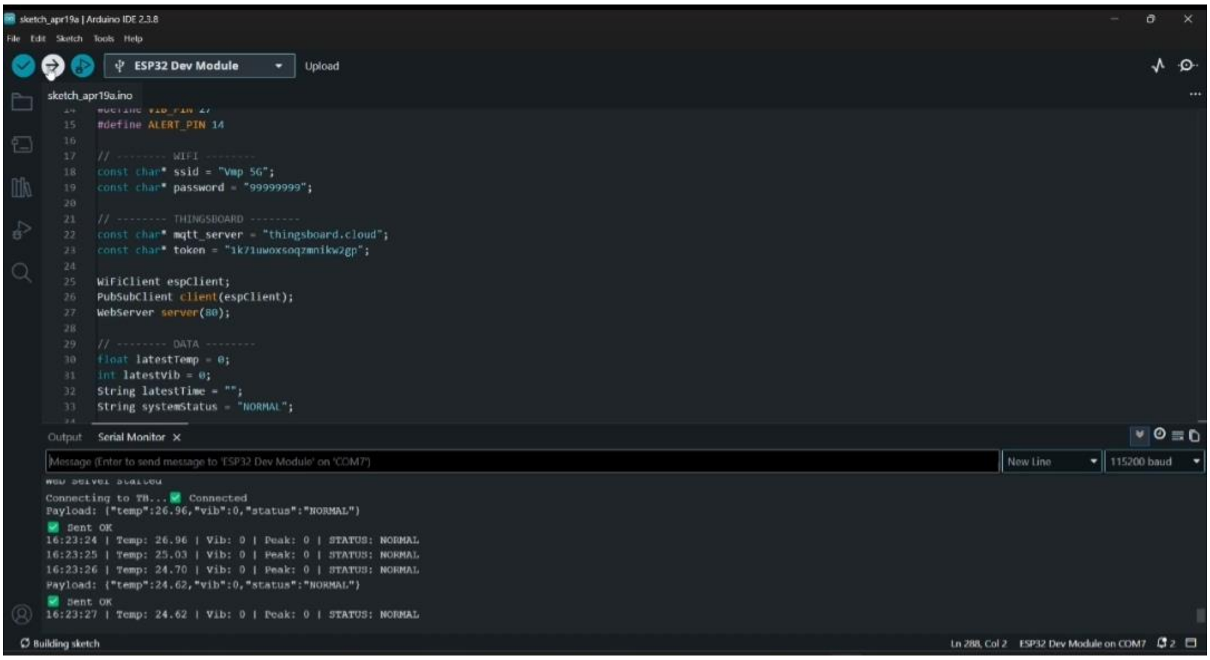

Figure 4.2: Serial Monitor Output Showing Real-Time Sensor Data and Status

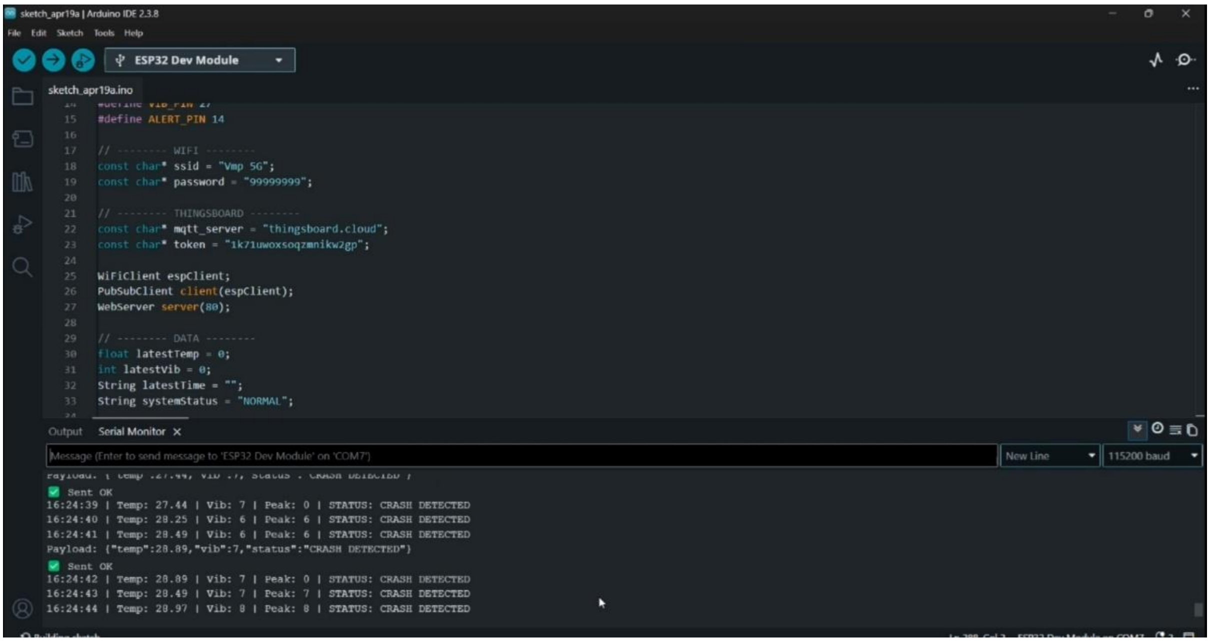

Figure 4.3: Serial Monitor Output During Crash Detection

4.3 Key Features

- Real-time monitoring
- Dual-condition crash detection
- Pre-crash data storage
- Automatic system restart
- Wireless vehicle control
- Cloud + Local dashboard

5.RESULTS AND PERFORMANCE ANALYSIS

5.1 Results

The implemented system successfully demonstrated:

- Accurate detection of crash events
- Continuous data logging in SD card
- Storage of pre-crash data using ring buffer
- Real-time visualization on cloud dashboard
- Functional local web interface
- Immediate motor shutdown during crash

5.2 Performance Analysis

Accuracy

- Dual-sensor logic significantly reduces false detection

Real-Time Performance

- Data updates every 1 second
- Low latency communication

System Stability

- Continuous operation without crash
- Auto-recovery using restart mechanism

5.3 Limitations

- Sensor precision is limited (low-cost sensors)
- Cloud monitoring depends on Wi-Fi
- No GPS tracking
- No camera recording

5.4 Limitations Identified During Testing

During the implementation and testing of the system, several practical limitations were observed:

- **Power Supply Instability:**
  The system initially faced issues with battery-based power delivery. Voltage fluctuations and insufficient current affected stable operation of the ESP32 and peripherals.

- **Buck Converter**(**Failure):**
  The DC-DC buck converter used for voltage regulation showed inconsistent performance and occasional failure, leading to unstable output voltage and system resets.

- **GPS Module Issues:**
  The GPS module failed to provide reliable data due to signal acquisition problems, especially in indoor environments. As a result, GPS functionality could not be fully integrated into the final system.

- **Communication Latency in Web Control:**

  A noticeable delay was observed between user input on the web dashboard and the corresponding action on the ESP32, affecting real-time responsiveness of vehicle control.

- **Limited Range and Weak Signal Strength:**

  When using Wi-Fi connectivity, the system experienced weak signal strength and reduced communication range. This impacted data transmission reliability.

- **Network Optimization Attempt:**

  To improve connectivity, the system was switched from standard Wi-Fi to mobile hotspot-based communication, which provided relatively better stability but still had range limitations.

CONCLUSION AND FUTURE SCOPE

The project successfully demonstrates the design and implementation of a **Mini Automotive Black Box system** using ESP32. The system integrates sensing, processing, storage, and communication into a unified embedded platform.

Key achievements include:

- Reliable crash detection using dual-condition logic
- Efficient pre-crash data storage using ring buffer
- Real-time monitoring through cloud and local interfaces
- Automatic safety response and system recovery

This project bridges the gap between theoretical concepts and practical implementation of automotive EDR systems.

Future Scope

Although the current system performs effectively, several enhancements can be considered in future work:

- Integration of GPS module for location tracking
- Addition of camera for visual recording
- Development of mobile application
- Use of high-precision industrial sensors
- AI-based crash prediction

These enhancements can further improve the system's intelligence, scalability, and real-world applicability.

References:

1. Y. Yao and E. M. Atkins, "The Smart Black Box: A Value-Driven Automotive Event Data Recorder," in *Proc. 2018 21st IEEE International Conference on Intelligent Transportation Systems (ITSC)*, Maui, HI, USA, Nov. 2018, pp. 973–978, doi: 10.1109/ITSC.2018.8569253.

2. C. Patil, Y. Marathe, K. Amoghimath, and S. David, "Low Cost Black Box for Cars," in *Proc. 2013 Texas Instruments India Educators' Conference (TIIEC)*, Bangalore, India, Apr. 2013, doi: 10.1109/TIIEC.2013.6757114.

3. K. Dhanam, P. Marichamy, and R. Mohan et al., "Event Data Recorder for Investigation, Legal Claim and Fault Analysis in Vehicles," *International Journal for Research Trends and Innovation (IJRTI)*, vol. 2, no. 4, 2017. *(No public DOI indexed — available at ijrti.org)*

4. "Affordable Black Box: A Smart Accident Detection System for Cars," in *Proc. 2021 IEEE 12th Control and System Graduate Research Colloquium (ICSGRC)*, Shah Alam, Malaysia, 2021, doi: 10.1109/ICSGRC53186.2021.9596106.

5. T. W. Chim, S. M. Yiu, C. Y. Yeung, V. O. K. Li, and L. C. K. Hui, "Secure, Privacy-Preserving, Distributed Motor Vehicle Event Data Recorder," in *Proc. 2013 International Conference on Connected Vehicles and Expo (ICCVE)*, Las Vegas, NV, USA, 2013, pp. 337–342, doi: 10.1109/ICCVE.2013.6799817.

6. *IEEE Standard for Motor Vehicle Event Data Recorder (MVEDR)*, IEEE Std 1616-2004, IEEE, 2005; revised as IEEE Std 1616-2021, doi: 10.1109/IEEESTD.2021.9654153.

7. "The Use of Event Data Recorder (EDR) – Black Box," *Annals of Spiru Haret University – Technical Series (ASTJ)*, 2014. *(No public DOI indexed)*

8. M. Thanzeel, N. Thomas B, S. Suresh, S. S. Krishna, and N. K. Nataraj, "Vehicle Black Box System," *International Journal of Science and Research Technology (IJSRT)*, vol. 2, no. 4, pp. 137–143, 2025, doi: 10.5281/zenodo.15179749.

9. Y. Wang, K. Hu, and Y. Zhou, "Wireless Bluetooth Car Collision Detection System," in *Proc. 2019 7th International Conference on Information, Communication and Networks (ICICN)*, Macao, China, Apr. 2019, doi: 10.1109/ICICN.2019.8834933.

10. J. Essien and C. Chimezie, "Ultrasonic Sensor-Based Embedded System for Vehicular Collision Detection and Alert," *Journal of Computer and Communications*, vol. 11, pp. 44–57, 2023, doi: 10.4236/jcc.2023.1110004.

11. "Embedded Based Vehicle Speed Moderation and Collisionless System," *International Journal for Research Trends and Innovation (IJRTI)*, ECE Dept., Udupi, India, vol. 2, 2017. *(No public DOI indexed — available at ijrti.org)*

12. M. A. Khan, T. Nawaz, U. S. Khan, A. Hamza, and N. Rashid, "IoT-Based Non-Intrusive Automated Driver Drowsiness Monitoring Framework for Logistics and Public Transport Applications to Enhance Road Safety," *IEEE Access*, vol. 11, pp. 14385–14397, Feb. 2023, doi: 10.1109/ACCESS.2023.3244008.

13. "Voice and Bluetooth Controlled Robot Car," *International Research Journal of Engineering and Technology (IRJET)*, vol. 6, no. 5, May 2019, pp. 1448–1451. *(No public DOI indexed — available at irjet.net)*

14. B. Swathi, S. Kamsali, K. Gayathri, and M. D. Lakshmi, "Design and Implementation of an IoT-Enabled Bluetooth Robot Car with Obstacle Avoidance," *International Journal of Intelligent Systems and Applications in Engineering (IJISAE)*, vol. 11, no. 6s, pp. 13–22, 2023. [Online]. Available: https://www.ijisae.org/index.php/IJISAE/article/view/2824.

15. "Embedded Based Vehicle Speed Moderation and Collisionless System," *International Journal for Research Trends and Innovation (IJRTI)*, ECE Dept., Udupi, India, vol. 2, 2017. *(No public DOI indexed)*

16. P. U. Prafulla, G. S. Yuvashree, Ramya Gowda, Rashmitha H. R., and Rajalakshmi, "Black Box with Intelligent Vehicle System," *International Journal of Engineering Research and Technology (IJERT)*, vol. 5, no. 18 (NConPCS-2017), 2017, doi: 10.17577/IJERTCONV5IS18015.

17. "Voice and Bluetooth Controlled Robot Car," *International Research Journal of Engineering and Technology (IRJET)*, vol. 6, no. 5, May 2019. *(No public DOI indexed)*

18. M. Vogel, "Event Data Recorder: The Black Box for Vehicles," DEKRA Technical Overview, DEKRA SE, Stuttgart, Germany, 2024. [Online]. Available: https://www.dekra.com/en/event-data-recorder-the-black-box-for-vehicles/.

19. National Academies of Sciences, Engineering, and Medicine, "Use of Event Data Recorder (EDR) Technology for Highway Crash Data Analysis," NCHRP Web-Only

Document 75, Transportation Research Board (TRB), Washington, DC, USA, 2005, doi: 10.17226/21974.

20. T. W. Chim, S. M. Yiu, C. Y. Yeung, V. O. K. Li, and L. C. K. Hui, "Secure, Privacy-Preserving, Distributed Motor Vehicle Event Data Recorder," in *Proc. 2013 International Conference on Connected Vehicles and Expo (ICCVE)*, Las Vegas, NV, USA, 2013, pp. 337–342, doi: 10.1109/ICCVE.2013.6799817.

21. J. Essien and C. Chimezie, "Ultrasonic Sensor-Based Embedded System for Vehicular Collision Detection and Alert," *Journal of Computer and Communications*, vol. 11, pp. 44–57, 2023, doi: 10.4236/jcc.2023.1110004.

22. M. A. Khan, T. Nawaz, U. S. Khan, A. Hamza, and N. Rashid, "IoT-Based Non-Intrusive Automated Driver Drowsiness Monitoring Framework for Logistics and Public Transport Applications to Enhance Road Safety," *IEEE Access*, vol. 11, pp. 14385–14397, Feb. 2023, doi: 10.1109/ACCESS.2023.3244008.

23. "Embedded Based Vehicle Speed Moderation and Collisionless System," *International Journal for Research Trends and Innovation (IJRTI)*, ECE Dept., Udupi, India, vol. 2, 2017. *(No public DOI indexed)*

24. B. Swathi, S. Kamsali, K. Gayathri, and M. D. Lakshmi, "Design and Implementation of an IoT-Enabled Bluetooth Robot Car with Obstacle Avoidance," *International Journal of Intelligent Systems and Applications in Engineering (IJISAE)*, vol. 11, no. 6s, pp. 13–22, 2023. [Online].

Available: https://www.ijisae.org/index.php/IJISAE/article/view/2824.